



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP3623
Database Programming 2025/2026
SEMESTER 1
Lab Assignment 2

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Section	:	02

PART A - DATA PREPARATION

A1 - Create Collections

- appointments

```
db.createCollection("appointments");
```

- loginSessions

```
db.createCollection("loginSessions");
```

A2 - Create Documents

- appointments

```
db.appointments.insertMany([
  {
    appointmentId: "A001",
    patientName: "Muhammad Aiman Hakim",
    doctor: "Dr. Siti Aisyah",
    department: "Cardiology",
    status: "COMPLETED",
    fee: 180,
    appointmentDate: ISODate("2025-01-05T09:30:00Z"),
  },
  {
    appointmentId: "A002",
    patientName: "Nur Ain Sofea",
    doctor: "Dr. Ahmad Firdaus",
    department: "Orthopedics",
    status: "PENDING",
    fee: 120,
    appointmentDate: ISODate("2025-01-08T10:00:00Z"),
  },
  {
    appointmentId: "A003",
    patientName: "Daniel Syafiq",
    doctor: "Dr. Nur Hidayah",
    department: "Dermatology",
    status: "COMPLETED",
    fee: 90,
    appointmentDate: ISODate("2025-01-03T08:00:00Z"),
  },
  {
    appointmentId: "A004",
    patientName: "Aisyah Nabila",
    doctor: "Dr. Faizal Rahman",
    department: "Neurology",
    status: "CANCELLED",
    fee: 250,
    appointmentDate: ISODate("2025-01-10T11:00:00Z"),
  },
])
```

- loginSessions

```
db.loginSessions.insertMany([
  {
    staffId: "S001",
    staffName: "Dr. Siti Aisyah",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.10",
    loginTime: ISODate("2025-01-05T08:10:00Z"),
  },
  {
    staffId: "S002",
    staffName: "Dr. Ahmad Firdaus",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.11",
    loginTime: ISODate("2025-01-08T09:00:00Z"),
  },
  {
    staffId: "S003",
    staffName: "Dr. Nur Hidayah",
    role: "Doctor",
    status: "INACTIVE",
    ipAddress: "192.168.1.12",
    loginTime: ISODate("2025-01-03T14:30:00Z"),
  },
  {
    staffId: "S004",
    staffName: "Dr. Faizal Rahman",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.13",
    loginTime: ISODate("2025-01-10T10:45:00Z"),
  },
])
```

PART B - APPOINTMENT REPORTING

B1 - Advanced Query & Sorting

```
db.appointments.find({status : "COMPLETED", fee: {$gte: 100, $lte: 300}},  
  {appointmentId: 1, patientName: 1, department: 1, fee: 1}).sort({department: 1, fee: -1})
```

Output:

```
[  
  {  
    "_id": {  
      "$oid": "696b4083212833dd30201e2b"  
    },  
    "appointmentId": "A001",  
    "patientName": "Muhammad Aiman Hakim",  
    "department": "Cardiology",  
    "fee": 180  
  },  
  {  
    "_id": {  
      "$oid": "696b4083212833dd30201e31"  
    },  
    "appointmentId": "A007",  
    "patientName": "Haziq Afiq",  
    "department": "Neurology",  
    "fee": 125  
  },  
  {  
    "_id": {  
      "$oid": "696b4083212833dd30201e33"  
    },  
    "appointmentId": "A009",  
    "patientName": "Muhammad Arif",  
    "department": "Psychiatry",  
    "fee": 200  
  }  
]
```

B2 - Aggregation Summary

```
db.appointments.aggregate([
  {$group: { _id: "$department", totalAppointments: {$sum: 1}, totalFees: {$sum: "$fee"} }},
  {$sort: {totalFees: -1}}
])
```

Output:

```
[
  {
    "_id": "Neurology",
    "totalAppointments": 2,
    "totalFees": 375
  },
  {
    "_id": "Psychiatry",
    "totalAppointments": 2,
    "totalFees": 350
  },
  {
    "_id": "Orthopedics",
    "totalAppointments": 3,
    "totalFees": 300
  },
  {
    "_id": "Cardiology",
    "totalAppointments": 1,
    "totalFees": 180
  },
  {
    "_id": "Dermatology",
    "totalAppointments": 1,
    "totalFees": 90
  },
  {
    "_id": "General Medicine",
    "totalAppointments": 1,
    "totalFees": 60
  }
]
```

C1 - Active Staff Sessions

```
db.loginSessions.find({status: "ACTIVE"},
    {staffName: 1, role: 1, ipAddress: 1, loginTime: 1}).sort({loginTime: -1})
```

Output:

```
[
{
  "_id": {
    "$oid": "696b4083212833dd30201e3d"
  },
  "staffName": "Ahmad Zulfikar",
  "role": "Administrator",
  "ipAddress": "192.168.1.16",
  "loginTime": {
    "$date": "2025-01-12T09:30:00Z"
  }
},
{
  "_id": {
    "$oid": "696b4083212833dd30201e38"
  },
  "staffName": "Dr. Faizal Rahman",
  "role": "Doctor",
  "ipAddress": "192.168.1.13",
  "loginTime": {
    "$date": "2025-01-10T10:45:00Z"
  }
},
{
  "_id": {
    "$oid": "696b4083212833dd30201e3b"
  },
  "staffName": "Ahmad Zulfikar",
  "role": "Receptionist",
  "ipAddress": "192.168.1.15",
  "loginTime": {
    "$date": "2025-01-09T08:00:00Z"
  }
},
{
```

C2 - Distinct and Count

a.

```
db.loginSessions.distinct("ipAddress", {status: "ACTIVE"})
```

Output:

```
[
  "192.168.1.10",
  "192.168.1.11",
  "192.168.1.13",
  "192.168.1.14",
  "192.168.1.15",
  "192.168.1.16"
]
```

b.

```
db.loginSessions.aggregate([
  {$match: {status: "ACTIVE"}},
  {$group: {_id: "$staffName", count: {$sum: 1}}}
])
```

Output:

```
[
  {
    "_id": "Ahmad Zulfikar",
    "count": 2
  },
  {
    "_id": "Dr. Faizal Rahman",
    "count": 1
  },
  {
    "_id": "Dr. Siti Aisyah",
    "count": 1
  },
  {
    "_id": "Dr. Ahmad Firdaus",
    "count": 1
  },
  {
    "_id": "Nur Aisyah",
    "count": 1
  },
  {
    "_id": "Dr. Farah Imani",
    "count": 1
  }
]
```

PART D – INDEXING & DATA MAINTENANCE

D1 – Compound Index

```
db.appointments.createIndex({department: 1, status: 1})

db.appointments.find({department: "Psychiatry", status: "COMPLETED"})
```

Output:

```
department_1_status_1

[
  {
    Edit Document
    "_id": {
      "$oid": "696b4083212833dd30201e33"
    },
    "appointmentId": "A009",
    "patientName": "Muhammad Arif",
    "doctor": "Dr. Nadia Karim",
    "department": "Psychiatry",
    "status": "COMPLETED",
    "fee": 200,
    "appointmentDate": {
      "$date": "2025-01-04T08:00:00Z"
    }
  }
]
```

D2 – Update & Delete Operations

a.

```
db.appointments.updateMany(
  {status: "CANCELLED"},
  {$set: {status: "ARCHIVED"}}
)
```

b.

```
db.loginSessions.deleteMany({status: "INACTIVE"})
```